Revision

$b = f(a)\ \ c = f(b)\ \Rightarrow\ c = f(a)$ *if a is changed , c will be chnaged accordingly)*

$$\frac{\partial c}{\partial a} = \frac{\partial c}{\partial b} * \frac{\partial b}{\partial a}$$ *Chain Rule*

$= f(\underline{W}) = w_1 . a_1 + w_2 . a_2 + w_3 . a_3 + \ldots\ldots + w_n . a_n = \sum_{i=1}^{n} w_i . a_i$

$$\frac{\partial b}{\partial w_1} = a_1 \quad \frac{\partial b}{\partial w_2} = a_2 \quad \frac{\partial b}{\partial w_i} = a_i$$

Conclusion: $b = f(\underline{W}) = \sum_{i=1}^{n} a_i w_i$ then $\frac{\partial b}{\partial w_i} = a_i$ *One Term ONLY*

$c = f(a_1), c = f(a_2), c = f(a_3)$ and $a_1 = f(b), a_2 = f(b), a_3 = f(b)$

$c = f(b) \Rightarrow$ if b is changed, c will be changed accordingly [*through* a_1, a_2, a_3]

$$\frac{\partial c}{\partial b} = \left[\frac{\partial c}{\partial a_1} * \frac{\partial a_1}{\partial b}\right] + \left[\frac{\partial c}{\partial a_2} * \frac{\partial a_2}{\partial b}\right] + \left[\frac{\partial c}{\partial a_3} * \frac{\partial a_3}{\partial b}\right] = \sum_{i=1}^{3} \frac{\partial c}{\partial a_i} * \frac{\partial a_i}{\partial b}$$